

# An Explicit 29-AND Circuit for the AES S-box

umizame

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## Abstract

We give an explicit straight-line program for the forward AES S-box using twenty-nine AND gates. The construction rests on Canright’s all-normal tower for the AES field and on the five-AND inversion of Boyar, Matthews, and Peralta in the field of sixteen elements. Its essential economy is a joint calculation over the field of four elements: the three pairwise products of three operands are obtained from three early ANDs and five late ANDs. Using this identity three times around the five-AND inversion gives the schedule  $9 + 5 + 15 = 29$ . Affine coordinate data then realize this schedule as a normalized XOR–AND graph and as the explicit straight-line program.

## 1 Introduction

The nonlinear step of the AES S-box is inversion in the field of 256 elements; the remaining operations are affine over  $\mathbb{F}_2$ . Canright’s all-normal tower makes this inversion especially economical by resolving it into arithmetic in successively smaller fields [1, Sec. 2]. Boyar, Matthews, and Peralta subsequently gave a five-AND circuit for the inversion which occurs at the middle level of that tower [2, Sec. 2.1, Fig. 2]. These two works supply, respectively, the tower coordinates and the middle-level inverse used below.

The straight-line program presented here uses the operations XOR, AND, and NOT, with unrestricted fan-out. An AND gate is multiplication in  $\mathbb{F}_2$ ; XOR and NOT are affine and therefore require no AND gate. The new ingredient is a schedule for three multiplications in  $\mathbb{F}_{16}$ . Their nine constituent multiplications in  $\mathbb{F}_4$  occur in three triples. In each triple the first product is needed before the  $\mathbb{F}_{16}$  inverse is known, whereas the other two are needed afterwards. A single identity computes the first product with three ANDs and completes the other two with five more. The nonlinear count is consequently

$$3 \cdot (3 + 5) + 5 = 29.$$

The proof proceeds entirely in Canright’s coordinates. Section 2 recalls the tower and the two multiplication formulae that will be used. Section 3 proves the joint  $\mathbb{F}_4$  identity. Section 4 places its three copies around the Boyar–Matthews–Peralta inverse. Section 5 supplies the affine coordinates that realize the construction as a normalized XOR–AND graph and as the explicit straight-line program.

## 2 Canright’s tower form

Let

$$\mathbb{F}_{256} = \mathbb{F}_2[t]/(t^8 + t^4 + t^3 + t + 1), \quad X = [t].$$

This is the polynomial representation specified in FIPS 197 [3, Secs. 3.2 and 4, Eqs. (4.1) and (4.3)]. For a byte  $x = \sum_{i=0}^7 b_i X^i$ , with  $b_i \in \mathbb{F}_2$ , set

$$U_j = b_{7-j} \quad (0 \leq j < 8),$$

so that  $U_0$  is the most significant input bit. If  $x^{-1} = \sum_{i=0}^7 v_i X^i$ , with  $v_i \in \mathbb{F}_2$  and  $0^{-1} = 0$ , FIPS 197 defines the output bits by

$$o_i = v_i + v_{i+4} + v_{i+5} + v_{i+6} + v_{i+7} + c_i, \quad \sum_{i=0}^7 c_i 2^i = 63, \quad (1)$$

where  $c_i \in \mathbb{F}_2$ , 63 is written in hexadecimal, and the subscripts on  $v_i$  are read modulo eight [3, Sec. 5.1.1, Eqs. (5.2)–(5.3)]. We use the same zero-inverse convention in every field below, and write the output in the most-significant-first order  $S_j = o_{7-j}$ .

We now take Canright's all-normal tower

$$\mathbb{F}_2 \subset \mathbb{F}_4 \subset \mathbb{F}_{16} \subset \mathbb{F}_{256},$$

with ordered bases for the three successive quadratic extensions

$$[W^2, W], \quad [Z^4, Z], \quad [Y^{16}, Y]$$

and defining relations

$$W^2 + W + 1 = 0, \quad Z^2 + Z + W^2 = 0, \quad Y^2 + Y + WZ = 0. \quad (2)$$

These are precisely the field and bases used in Canright's all-normal implementation [1, Sec. 2.1]. The relations give

$$W^3 = 1, \quad Z^4 + Z = 1, \quad Z^5 = W^2, \quad Y^{16} + Y = 1, \quad Y^{17} = WZ. \quad (3)$$

Indeed, squaring the middle relation gives  $Z^4 = Z + W^2 + W = Z + 1$ , and multiplication by  $Z$  then gives  $Z^5 = W^2$ . At the upper level, the two roots of the quadratic in (2) are  $Y$  and  $Y + 1$ ; the nontrivial  $\mathbb{F}_{16}$ -conjugate of  $Y$  is  $Y^{16}$ . Their sum and product give the last two identities.

Canright's change-of-basis matrix maps the tower coordinates to the AES polynomial coordinates [1, Sec. 4, Eq. (10)]. Its inverse carries the FIPS byte to

$$G = G_1 Y^{16} + G_0 Y, \quad G_0, G_1 \in \mathbb{F}_{16}.$$

The rows of this inverse map, in the bit order just fixed, are recorded in Appendix A. For the formulas below, it is convenient to multiply each  $\mathbb{F}_4$ -coordinate by  $W$  and put

$$H = WG = LY^{16} + RY, \quad L, R \in \mathbb{F}_{16}. \quad (4)$$

This scaling is affine, since for  $a_0, a_1 \in \mathbb{F}_2$ ,

$$W(a_0 W^2 + a_1 W) = (a_0 + a_1) W^2 + a_0 W. \quad (5)$$

**Lemma 1** (The upper norm). *For  $H$  as in (4), define*

$$D = LR + WZ(L + R)^2, \quad Q = D^{-1}. \quad (6)$$

*Then*

$$H^{-1} = RQY^{16} + QLY. \quad (7)$$

*Consequently  $WH^{-1} = G^{-1}$ .*

*Proof.* The sixteenth-power Frobenius fixes  $\mathbb{F}_{16}$  and exchanges  $Y^{16}$  and  $Y$ . Thus

$$H^{16} = RY^{16} + LY.$$

Using  $Y^{16} + Y = 1$  and  $Y^{17} = WZ$ , multiplication gives

$$HH^{16} = LR + WZ(L^2 + R^2) = LR + WZ(L + R)^2 = D.$$

If  $H \neq 0$ , division by this norm proves (7). If  $H = 0$ , the basis  $[Y^{16}, Y]$  gives  $L = R = 0$ , whence  $D = Q = 0$ , and the same formula holds under the declared convention. Finally,  $H = WG$  and  $W^3 = 1$ , so  $WH^{-1} = G^{-1}$ .  $\square$

The three products  $LR$ ,  $RQ$ , and  $QL$  will be formed with the following version of Canright's normal-basis multiplier [1, Sec. 2.1, Fig. 3].

**Lemma 2** (Multiplication in  $\mathbb{F}_{16}$ ). *Let*

$$A = A_1Z^4 + A_0Z, \quad B = B_1Z^4 + B_0Z, \quad A_i, B_i \in \mathbb{F}_4,$$

and set

$$\ell = A_0B_0, \quad h = (WA_1)(WB_1), \quad s = (A_0 + WA_1)(B_0 + WB_1). \quad (8)$$

Then

$$AB = (\ell + h + Ws)Z^4 + (W^2h + Ws)Z. \quad (9)$$

*Proof.* The relations in (3) give

$$\begin{aligned} (Z^4)^2 &= WZ^4 + W^2Z, \\ Z^2 &= W^2Z^4 + WZ, \\ Z^5 &= W^2Z^4 + W^2Z. \end{aligned}$$

Hence, writing  $C = A_1B_0 + A_0B_1$ , direct multiplication gives

$$\begin{aligned} AB &= (WA_1B_1 + W^2A_0B_0 + W^2C)Z^4 \\ &\quad + (W^2A_1B_1 + WA_0B_0 + W^2C)Z. \end{aligned}$$

Now  $h = W^2A_1B_1$  and  $s = \ell + h + WC$ . Substitution, together with  $1 + W = W^2$ , reduces the two coefficients to those in (9).  $\square$

### 3 Three products in $\mathbb{F}_4$

Write

$$\tau(x) = x + x^2 \quad (x \in \mathbb{F}_4)$$

for the trace to  $\mathbb{F}_2$ . The normal basis makes both coordinates visible as traces.

**Lemma 3.** *If  $x = x_0W^2 + x_1W$ , with  $x_0, x_1 \in \mathbb{F}_2$ , then*

$$\tau(x) = x_0 + x_1, \quad \tau(Wx) = x_1, \quad \tau(W^2x) = x_0. \quad (10)$$

For all  $x, y \in \mathbb{F}_4$ ,

$$\tau(x)\tau(Wy) + \tau(Wx)\tau(y) = \tau(xy^2), \quad (11)$$

$$\tau(x)\tau(y) + \tau(Wx)\tau(Wy) = \tau(Wxy). \quad (12)$$

*Proof.* Squaring  $x_0W^2 + x_1W$  exchanges its two normal-basis coordinates, which proves (10). If  $y = y_0W^2 + y_1W$ , with  $y_0, y_1 \in \mathbb{F}_2$ , the left side of (11) is  $x_0y_1 + x_1y_0$ , while the left side of (12) is  $x_0y_0 + x_0y_1 + x_1y_0$ . Multiplication with  $W^2 + W = 1$  shows that these are respectively  $\tau(xy^2)$  and  $\tau(Wxy)$ .  $\square$

Let  $a, b, c \in \mathbb{F}_4$ . We first recover  $ab$  with three ANDs that do not depend on  $c$ ; five later ANDs will then recover  $bc$  and  $ca$  jointly. The three early products are

$$p_1 = \tau(a)\tau(b), \quad p_2 = \tau(Wa)\tau(Wb), \quad p_3 = \tau(W^2a)\tau(W^2b). \quad (13)$$

Since  $\tau(W^2x) = \tau(x) + \tau(Wx)$ , equations (11) and (12) give

$$\sigma := p_1 + p_2 + p_3 = \tau(ab^2), \quad ab = (p_1 + p_3)W^2 + (p_1 + p_2)W. \quad (14)$$

For the second formula, the coefficient of  $W$  is  $\tau(Wab) = p_1 + p_2$ ; applying (12) after cycling both arguments by  $W^2$  gives the coefficient of  $W^2$  as  $\tau(W^2ab) = p_1 + p_3$ .

For distinct  $r, t \in \{1, W, W^2\}$ , put

$$e_{r,t} = (\tau(ra) + \tau(tb) + \tau(Wc))(\tau(Wra) + \tau(Wtb) + \tau(c)). \quad (15)$$

Among the six off-diagonal cells  $e_{r,t}$ , the five  $e$ -terms used below form the union of the rows  $r = W, W^2$  and the columns  $t = 1, W^2$ .

**Proposition 4** (The five late products). *In addition to  $p_1, p_2, p_3$ , define*

$$\begin{aligned} p_4 &= e_{1,W^2}, & p_5 &= e_{W,W^2}, \\ p_6 &= e_{W^2,1} + \tau(a + Wb + c), & p_7 &= e_{W,1}, \\ p_8 &= e_{W^2,W} + \tau(a + W^2b + c). \end{aligned} \quad (16)$$

Each  $p_i$  in (16) is the output of one AND gate. Moreover,

$$\tau(W^2bc) = \sigma + p_5 + p_7 + \tau(b), \quad (17)$$

$$\tau(bc) = \sigma + p_6 + p_8 + \tau(W^2b), \quad (18)$$

$$\tau(Wca) = \sigma + p_6 + p_7 + \tau(a + Wb + c) + \tau(W^2a), \quad (19)$$

$$\tau(ca) = \sigma + p_4 + p_5 + \tau(Wa). \quad (20)$$

Thus  $ab$ ,  $bc$ , and  $ca$  are obtained from eight AND gates, the first three of which do not depend on  $c$ .

*Proof.* We first compute the sum of two cells in a common row and in a common column; the stated formulas then follow after accounting for the two affine shears. Write  $\tau_r(x) = \tau(rx)$  and  $\eta(x) = \tau(x)\tau(Wx)$ . The three bits  $\tau_1(x), \tau_W(x), \tau_{W^2}(x)$  sum to zero. If  $\{r, t, u\} = \{1, W, W^2\}$ , expansion of the two cells in a fixed row gives

$$\begin{aligned} e_{r,t} + e_{r,u} &= \tau_r(a)\tau_{Wr}(b) + \tau_{Wr}(a)\tau_r(b) \\ &\quad + \tau_W(c)\tau_{Wr}(b) + \tau_1(c)\tau_r(b) \\ &\quad + \tau_t(b)\tau_{Wt}(b) + \tau_u(b)\tau_{Wu}(b). \end{aligned} \quad (21)$$

The first line is  $\tau(ab^2) = \sigma$  by (11), since  $r^3 = 1$ . The second is  $\tau(Wrbc)$  by (12). The last line is  $\eta(tb) + \eta(ub)$ . Writing  $x = x_0W^2 + x_1W$ , with  $x_0, x_1 \in \mathbb{F}_2$ , gives

$$\eta(x) = x_1(x_0 + x_1) = x_0x_1 + x_1 = x^3 + \tau(W^2x).$$

Here  $x^3$  is the norm from  $\mathbb{F}_4$  to  $\mathbb{F}_2$ . Since  $t^3 = u^3 = 1$ , the two norm terms cancel; since  $t + u = r$ , the remaining trace is  $\tau(W^2rb)$ . Thus

$$e_{r,t} + e_{r,u} = \sigma + \tau(Wrbc) + \tau(W^2rb). \quad (22)$$

The same argument down a column gives

$$\begin{aligned} e_{r,t} + e_{u,t} &= \tau_t(b)\tau_{Wt}(a) + \tau_{Wt}(b)\tau_t(a) \\ &\quad + \tau_W(c)\tau_{Wt}(a) + \tau_1(c)\tau_t(a) \\ &\quad + \tau_r(a)\tau_{Wr}(a) + \tau_u(a)\tau_{Wu}(a), \end{aligned}$$

and therefore

$$e_{r,t} + e_{u,t} = \sigma + \tau(Wtca) + \tau(W^2ta). \quad (23)$$

Here the first line is again  $\sigma$  by (11); explicitly,  $\tau(ba^2) = \tau((ba^2)^2) = \tau(ab^2)$ , by Frobenius invariance of the trace. The second line follows from (12). The last line is  $\eta(ra) + \eta(ua)$ ; cancellation of the two norms and  $r + u = t$  leave  $\tau(W^2ta)$ .

Equations (22) and (23) organize the reconstruction: row sums give traces of  $bc$ , whereas column sums give traces of  $ca$ .

It remains to account for the two sheared cells. Write the two factors of  $e_{W^2,1}$  as  $f$  and  $g$ . The second is  $g = \tau(a + Wb + c)$ ; hence

$$p_6 = fg + g = (f + g)g,$$

because  $g^2 = g$  in  $\mathbb{F}_2$ . Thus  $p_6$  is one AND, with factors  $f + g$  and  $g$ . The second factor of  $e_{W^2,W}$  is  $\tau(a + W^2b + c)$ , and the same argument proves the assertion for  $p_8$ .

Taking  $r = W$  and columns  $W^2, 1$  in (22) gives (17). Taking  $r = W^2$  and columns  $1, W$ , and observing that the two shear terms sum to  $\tau(b)$ , while  $\tau(Wb) + \tau(b) = \tau(W^2b)$ , gives (18). Taking column  $1$  and rows  $W^2, W$  in (23) gives (19); taking column  $W^2$  and rows  $1, W$  gives (20).

Finally, (10) gives the explicit reconstructions

$$\begin{aligned} bc &= \tau(W^2bc)W^2 + (\tau(bc) + \tau(W^2bc))W, \\ ca &= (\tau(ca) + \tau(Wca))W^2 + \tau(Wca)W. \end{aligned}$$

Thus the four displayed traces recover  $bc$  and  $ca$ , while (14) has already recovered  $ab$ .  $\square$

## 4 The twenty-nine AND gates

Write

$$L = L_1Z^4 + L_0Z, \quad R = R_1Z^4 + R_0Z, \quad Q = Q_1Z^4 + Q_0Z, \quad L_i, R_i, Q_i \in \mathbb{F}_4.$$

Apply Proposition 4 to the three triples

$$\begin{array}{ccc} a & b & c \\ \hline L_0 & R_0 & Q_0 \\ WL_1 & WR_1 & WQ_1 \\ L_0 + WL_1 & R_0 + WR_1 & Q_0 + WQ_1. \end{array} \tag{24}$$

In the three rows, the products  $ab$  are respectively the quantities  $\ell, h, s$  of Lemma 2 for  $LR$ . The products  $bc$  are the same three quantities for  $RQ$ , and the products  $ca$  are those for  $QL$ . Thus the first three ANDs in each row make  $LR$  available; the final five, once  $Q$  is known, complete  $RQ$  and  $QL$ .

It remains to obtain  $Q = D^{-1}$  between these two stages. We use the five-product inversion of Boyar, Matthews, and Peralta, written in Canright's coordinates.

**Proposition 5** (The middle inverse). *Write*

$$D = (d_0W^2 + d_1W)Z^4 + (d_2W^2 + d_3W)Z. \tag{25}$$

Here  $d_0, \dots, d_3 \in \mathbb{F}_2$ . Define, in order,

$$\begin{aligned} m_1 &= (d_0 + d_1)(d_2 + d_3), \\ m_2 &= d_1(d_2 + m_1), \\ m_3 &= (d_0 + m_1)d_3, \\ m_4 &= d_2(m_1 + m_3), \\ m_5 &= (d_0 + m_2)(d_2 + m_1 + m_4). \end{aligned} \tag{26}$$

Also define

$$\begin{aligned} q_0 &= d_2 + d_3 + m_4, & q_1 &= d_3 + m_3 + m_4, \\ q_2 &= d_0 + d_1 + m_2 + m_5, & q_3 &= d_1 + m_5. \end{aligned} \tag{27}$$

Then the five products in (26) give

$$Q = (q_0 W^2 + q_1 W) Z^4 + (q_2 W^2 + q_3 W) Z = D^{-1}.$$

*Proof.* Boyar, Matthews, and Peralta represent  $\Delta$  using bits  $x_1, \dots, x_4 \in \mathbb{F}_2$  as

$$\Delta = (x_1 W + x_2 W^2) Z^2 + (x_3 W + x_4 W^2) Z^8$$

and give a five-AND circuit for  $\Delta^{-1}$  [2, Sec. 2.1, Fig. 2]. Set

$$(x_1, x_2, x_3, x_4) = (d_0 + d_1, d_0, d_2 + d_3, d_2). \tag{28}$$

Since  $Z^{16} = Z$ , squaring  $\Delta$  and multiplying by  $W^2$  gives  $D = W^2 \Delta^2$ . Indeed,  $W^2((d_0 + d_1)W + d_0 W^2)^2 = d_0 W^2 + d_1 W$ , and the same identity holds for the lower pair. Taking inverses and using  $W^3 = 1$  then gives

$$D = W^2 \Delta^2, \quad Q = D^{-1} = W(\Delta^{-1})^2. \tag{29}$$

Under (28), the five ANDs in that circuit are exactly  $m_1, \dots, m_5$  in (26). Its four output coordinates are

$$\begin{aligned} y_1 &= d_3 + m_3 + m_4, & y_2 &= d_2 + m_3, \\ y_3 &= d_1 + m_5, & y_4 &= d_0 + m_2, \end{aligned}$$

where

$$\Delta^{-1} = (y_1 W + y_2 W^2) Z^2 + (y_3 W + y_4 W^2) Z^8.$$

For  $\alpha = y_1 W + y_2 W^2$ , one has

$$W\alpha^2 = (y_1 + y_2)W^2 + y_1 W.$$

Applying this to both pairs in (29) yields precisely the four expressions in (27). The five-product circuit returns zero on zero, so the identity also respects the convention fixed above.  $\square$

We may now read off the entire schedule. The three early gates in each row of (24) use only  $L$  and  $R$ , and nine ANDs therefore produce  $LR$ . Squaring and multiplication by the constant  $WZ$  are linear, so  $D = LR + WZ(L + R)^2$  is then available without another AND. Proposition 5 produces  $Q$  with five ANDs. The five late gates in each row of (24) then produce  $RQ$  and  $QL$  with fifteen further ANDs. Lemma 1 gives  $H^{-1}$ . Multiplication by  $W$ , the combined linear map from the tower coordinates to the linear part of the FIPS output, and addition of the FIPS constant are all affine. This schedule is acyclic, and its nonlinear tally is

$$9 + 5 + 15 = 29.$$

## 5 The explicit circuit

To separate the nonlinear schedule from a particular affine XOR network, Appendix B records the construction as a normalized XOR–AND graph (XAG). This XAG uses the ordered affine basis

$$(1, U_0, \dots, U_7, A_1, \dots, A_{29}).$$

For an eight-bit linear mask  $\mu = \sum_{j=0}^7 \mu_j 2^j$ , which is written in hexadecimal when displayed, define

$$\lambda_\mu = \sum_{j=0}^7 \mu_j U_j;$$

thus bit  $j$  selects  $U_j$ , and the least significant bit selects  $U_0$ . By contrast, the XAG uses affine masks with an additional least-significant constant bit. In an AND row for  $A_i$ , bit zero selects the constant, bits one through eight select  $U_0, \dots, U_7$ , and bit  $8 + j$  selects  $A_j$  for  $1 \leq j < i$ . The product of the two affine forms is  $A_i$ . An output row is an affine mask in the complete basis.

Appendix A gives all coordinate data needed to express the preceding construction in these masks, together with the gate ranges for the three rows of (24). The five products  $m_1, \dots, m_5$  are  $A_{10}, \dots, A_{14}$ .

To illustrate the substitution, the lower row has

$$L_0 = \lambda_{41} W^2 + \lambda_{c6} W, \quad R_0 = \lambda_{74} W^2 + \lambda_{86} W.$$

Encoding these forms in the affine-mask convention above gives the three early factor pairs, in the orientation used in Appendix B,

$$(1e4, 10e), \quad (10c, 18c), \quad (e8, 82).$$

These are the first three rows of Appendix B. The same substitution gives  $A_4$  through  $A_9$ . The four masks of  $D$  in Appendix A give

$$(d_0 + d_1, d_2 + d_3) = (1ecf6, 1d06a),$$

the factor pair for  $A_{10}$ ; successive substitution in (26) gives  $A_{11}$  through  $A_{14}$ .

Substituting the four masks of  $Q$  into (16) in each of the three rows gives the fifteen late rows in the order tabulated in Appendix A. Finally, Lemma 2 reconstructs  $RQ$  and  $QL$ , and the affine rows in Appendix A express their coordinates as the eight output masks printed in Appendix B. Thus every row of Appendix B is obtained from the displayed formulae by successive XOR substitution. A concrete affine realization of this XAG is the straight-line program supplied in the repository:

[circuits/aes-sbox-fwd-g228-a29-d35-ad6.slp](#).

Scanning its assignments in order expresses every wire as an affine form in the inputs, the constant, and the outputs of preceding AND gates. XOR and NOT update the affine mask; each AND assignment records its two factor masks and introduces the next  $A_i$ . Consequently the program’s AND assignments give, in order, the twenty-nine factor rows in Appendix B, and its final affine forms give the eight output rows.

**Theorem 6.** *The straight-line program above computes the forward AES S-box specified in FIPS 197 and contains twenty-nine AND gates.*

*Proof.* The inverse of Canright’s change-of-basis matrix sends the AES polynomial coordinates to  $G$ , and (4) gives  $H = WG$ . Lemma 2, Proposition 4, and Proposition 5 construct  $RQ$  and  $QL$  with twenty-nine AND gates. Lemma 1 then gives  $WH^{-1} = G^{-1}$ , including at zero. The affine coordinates in Appendix A apply multiplication by  $W$ , Canright’s tower-to-polynomial map, and the linear part of the FIPS affine map, before adding the constant in (1). They therefore give the FIPS output in the declared bit order and the normalized XAG in Appendix B. The straight-line program above is a concrete affine realization of that XAG, so it computes the same output; its twenty-nine AND assignments are precisely the twenty-nine AND gates of the construction.  $\square$

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## A Affine coordinates

We use both hexadecimal-mask conventions of Section 5. On the ordered linear input basis  $(U_0, \dots, U_7)$ , bit  $j$  selects  $U_j$ . On an ordered affine basis  $(1, V_0, \dots, V_{n-1})$ , bit zero selects the constant and bit  $j + 1$  selects  $V_j$ . All masks below are written in hexadecimal.

The rows of the inverse map from AES polynomial coordinates to Canright’s tower coordinates, and the corresponding rows after the scaling  $H = WG$ , are as follows [1, Sec. 4, Eq. (10)]. The coordinate order is the tensor order induced by  $[Y^{16}, Y]$ ,  $[Z^4, Z]$ , and  $[W^2, W]$ .

| coordinate     | $G$ | $H = WG$ |
|----------------|-----|----------|
| $Y^{16}Z^4W^2$ | e7  | 69       |
| $Y^{16}Z^4W$   | 8e  | e7       |
| $Y^{16}ZW^2$   | c6  | 41       |
| $Y^{16}ZW$     | 87  | c6       |
| $YZ^4W^2$      | d9  | 59       |
| $YZ^4W$        | 80  | d9       |
| $YZW^2$        | 86  | 74       |
| $YZW$          | f2  | 86       |

In particular,

$$\begin{aligned} L_1 &= \lambda_{69}W^2 + \lambda_{e7}W, & L_0 &= \lambda_{41}W^2 + \lambda_{c6}W, \\ R_1 &= \lambda_{59}W^2 + \lambda_{d9}W, & R_0 &= \lambda_{74}W^2 + \lambda_{86}W. \end{aligned}$$

The three rows of (24), with their gate ranges, are

| row          | $a$          | $b$          | $c$          | early; late                              |
|--------------|--------------|--------------|--------------|------------------------------------------|
| lower        | $L_0$        | $R_0$        | $Q_0$        | $A_1\text{--}A_3; A_{15}\text{--}A_{19}$ |
| scaled upper | $WL_1$       | $WR_1$       | $WQ_1$       | $A_4\text{--}A_6; A_{20}\text{--}A_{24}$ |
| sum          | $L_0 + WL_1$ | $R_0 + WR_1$ | $Q_0 + WQ_1$ | $A_7\text{--}A_9; A_{25}\text{--}A_{29}$ |

The input masks for  $(a, b)$  in these rows, as pairs of  $[W^2, W]$ -coordinates, are

$$(41, c6), (74, 86); \quad (8e, 69), (80, 59); \quad (cf, af), (f4, df).$$

Within every late range, the gates are  $p_4, p_5, p_6, p_7, p_8$  in that order;  $p_6$  and  $p_8$  use the sheared factors displayed in the proof of Proposition 4.

On the affine basis  $(1, U_0, \dots, U_7, A_1, \dots, A_9)$ , the coordinate masks of  $D$  in (25) are

| coordinate | mask  |
|------------|-------|
| $d_0$      | 35afc |
| $d_1$      | 2b60a |
| $d_2$      | 330ea |
| $d_3$      | 2e080 |

On the basis enlarged through  $A_{14}$ , the four coordinate masks of  $Q$  are

| coordinate | mask   |
|------------|--------|
| $q_0$      | 21d06a |
| $q_1$      | 32e080 |
| $q_2$      | 49ecf6 |
| $q_3$      | 42b60a |

For the output coordinates, write

$$RQ = (T_0W^2 + T_1W)Z^4 + (T_2W^2 + T_3W)Z,$$

$$QL = (T_4W^2 + T_5W)Z^4 + (T_6W^2 + T_7W)Z.$$

Here  $T_0, \dots, T_7 \in \mathbb{F}_2$ . In the same most-significant-first convention, composing Canright's tower-to-polynomial map [1, Sec. 4, Eq. (10)] with the linear part of the FIPS affine map [3, Sec. 5.1.1, Eq. (5.4)] gives the rows

$$(14, 11, 82, 15, 1f, b6, 4c, 4a).$$

Composing multiplication by  $W$  with this linear map and then adding the constant 63 gives the following affine masks on  $(1, T_0, \dots, T_7)$ :

| $S_0$ | $S_1$ | $S_2$ | $S_3$ | $S_4$ | $S_5$ | $S_6$ | $S_7$ |
|-------|-------|-------|-------|-------|-------|-------|-------|
| 078   | 067   | 083   | 07e   | 074   | 0da   | 191   | 18b   |

After expressing the  $T_i$  on the complete XAG basis, these become

| output | mask       | output | mask       |
|--------|------------|--------|------------|
| $S_0$  | 2491ba35ce | $S_1$  | 257bba3b95 |
| $S_2$  | 2e6532ee01 | $S_3$  | 19dbba4b98 |
| $S_4$  | 189eba357c | $S_5$  | c0332e152  |
| $S_6$  | db00b2bf9  | $S_7$  | 19b508a44b |

## B The normalized XAG

Listing 1: The normalized XAG `certificates/aes29.xag`.

```
AES29-XAG 1
AND_COUNT 29
AND A01 1e4 10e
AND A02 10c 18c
AND A03 e8 82
AND A04 1b2 1ce
AND A05 b2 d2
AND A06 100 11c
AND A07 56 c0
AND A08 1be 15e
AND A09 1e8 19e
```

```
AND A10 1ecf6 1d06a
AND A11 2b60a 730ea
AND A12 75afc 2e080
AND A13 330ea 140000
AND A14 b5afc 2730ea
AND A15 42b7ec b5a94
AND A16 42b76e b5b9a
AND A17 49ed92 b5afe
AND A18 42b662 b5b72
AND A19 49ec9e b5b1a
AND A20 21d0a4 32e1e0
AND A21 21d1b8 32e02e
AND A22 133138 32e1fc
AND A23 21d10a 32e12e
AND A24 13318a 32e04e
AND A25 636748 39bb74
AND A26 6366d6 39bbb4
AND A27 5adcaa 39bb02
AND A28 636768 39ba5c
AND A29 5add14 39bb54
OUTPUT_COUNT 8
OUTPUT S0 2491ba35ce
OUTPUT S1 257bba3b95
OUTPUT S2 2e6532ee01
OUTPUT S3 19dbba4b98
OUTPUT S4 189eba357c
OUTPUT S5 c0332e152
OUTPUT S6 db00b2bf9
OUTPUT S7 19b508a44b
END
```

## References

- [1] D. Canright, “A Very Compact S-box for AES,” in *Cryptographic Hardware and Embedded Systems – CHES 2005*, Lecture Notes in Computer Science, vol. 3659, pp. 441–455, Springer, 2005. [doi:10.1007/11545262\\_32](https://doi.org/10.1007/11545262_32).
- [2] J. Boyar, P. Matthews, and R. Peralta, “Logic Minimization Techniques with Applications to Cryptology,” *Journal of Cryptology*, vol. 26, pp. 280–312, 2013. [doi:10.1007/s00145-012-9124-7](https://doi.org/10.1007/s00145-012-9124-7).
- [3] National Institute of Standards and Technology, *Advanced Encryption Standard (AES)*, NIST FIPS 197-upd1, originally published 2001, updated 9 May 2023. [doi:10.6028/NIST.FIPS.197-upd1](https://doi.org/10.6028/NIST.FIPS.197-upd1).